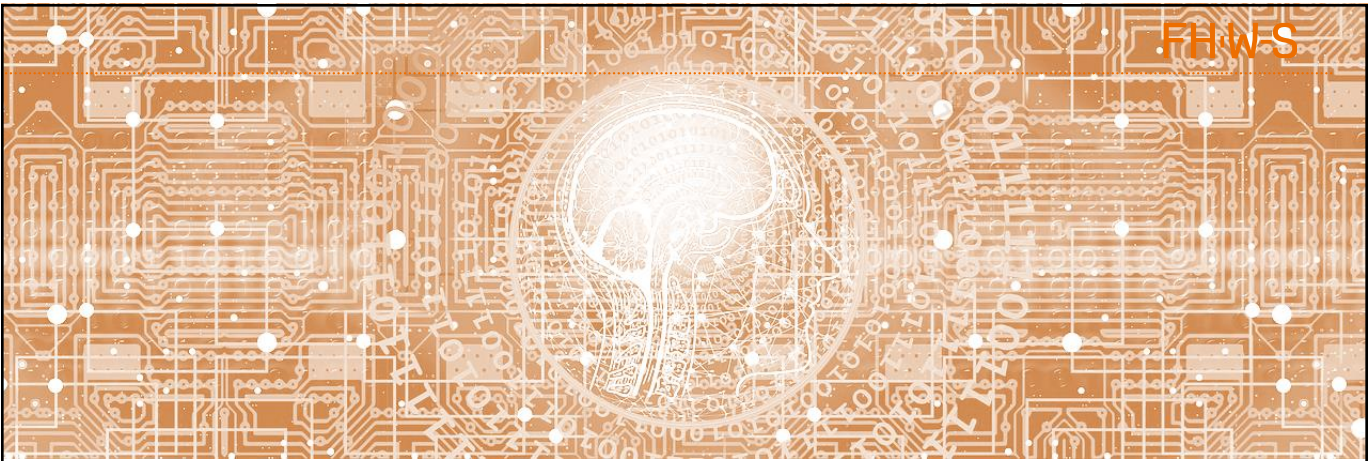




Software Engineering and Cyber Security

Prof. Dr. Marian Daun

FHWS University of Applied Sciences
Würzburg-Schweinfurt



Group Discussion

Expectations

FHWS University of Applied Sciences
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2

Skills a Software Engineer Needs

Team work / Social competence

Technical competence

- Talking competence (Talking about an Architecture, Requirements, Code, Test cases, etc.)
- Presentation skills
- Ability to learn and google
- Writing code

Tool competence

- Powerpoint
- Other MS-Office products
- GIT and SVN
- IDEs (e.g., Eclipse, IntelliJ)
- Advanced Tools (MATLAB/SIMULINK, DOORS, model editors, etc.)

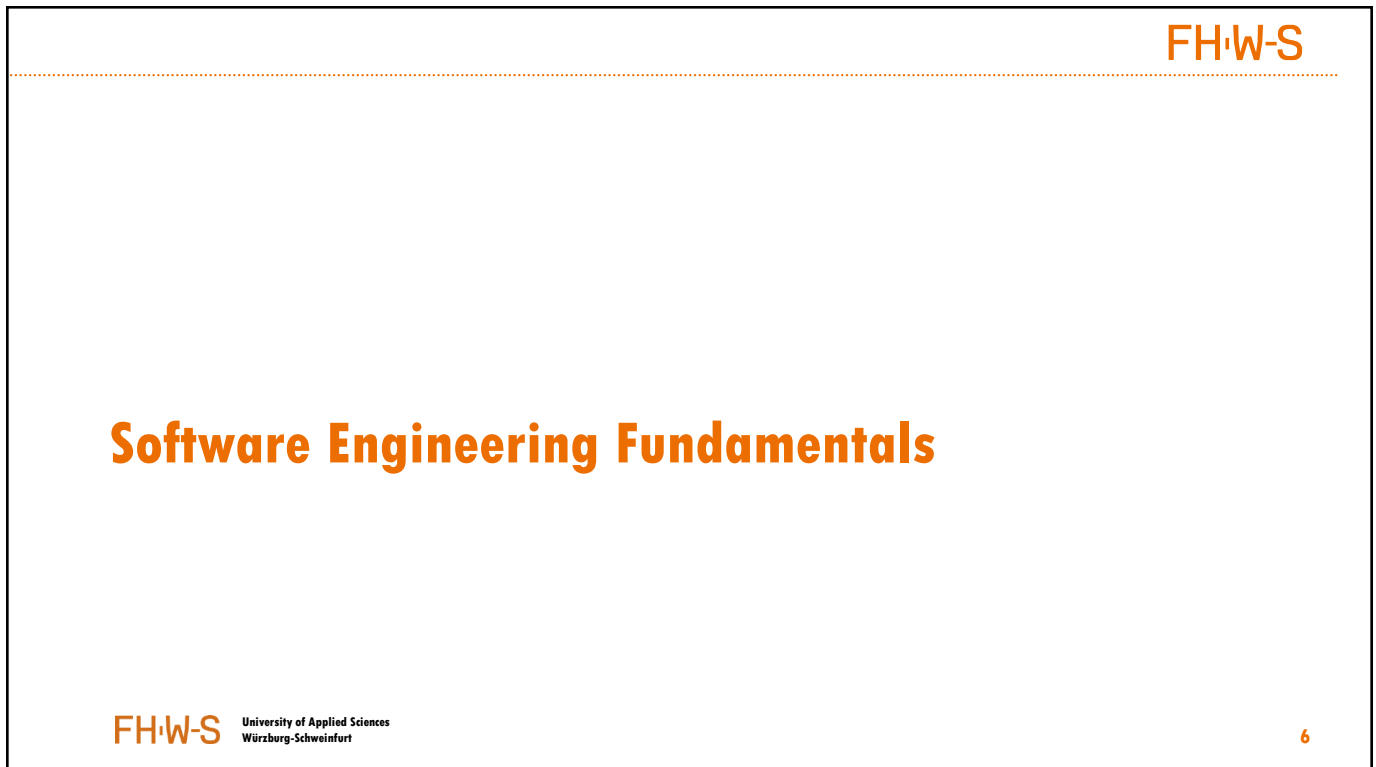
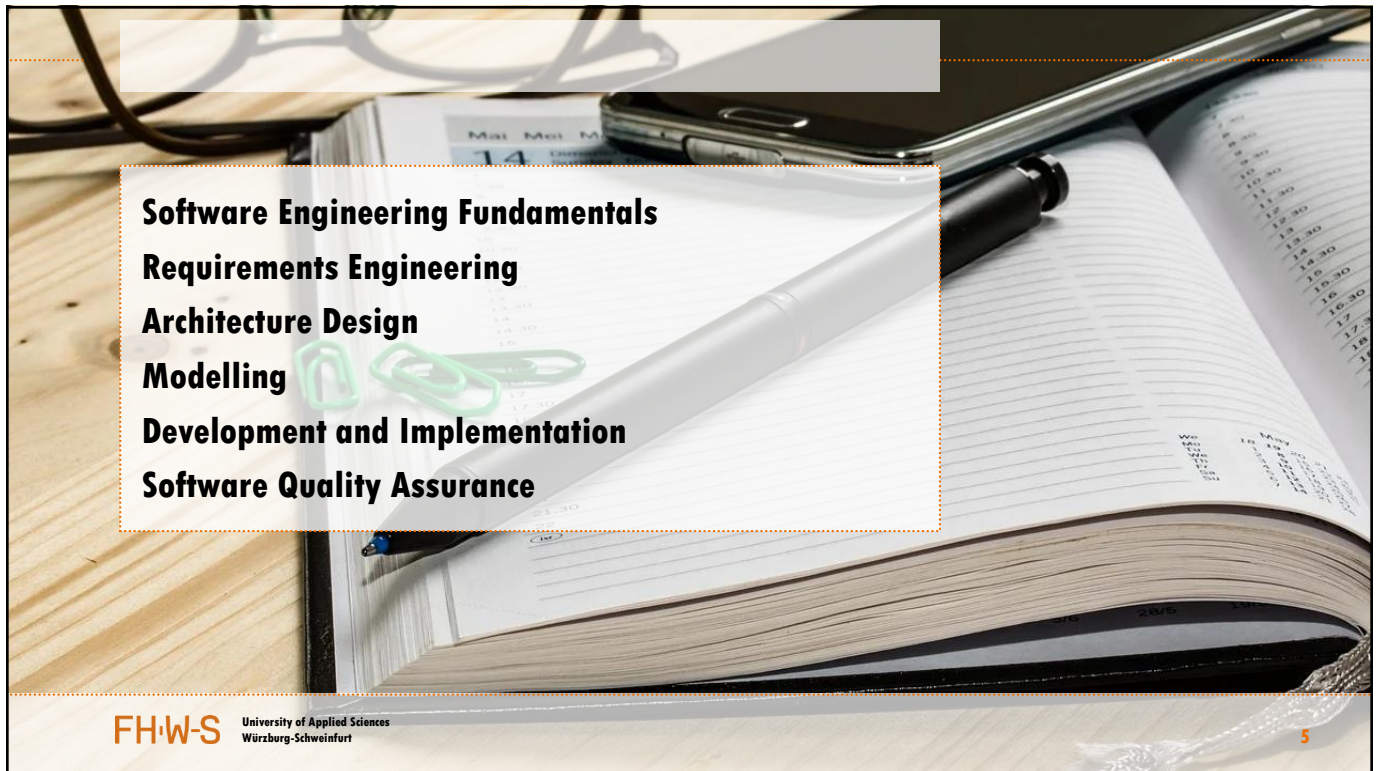
Literature

Important literature given on lecture slides

Must-read literature linked in the moodle course

If you need a comprehensive book:

Ian Sommerville: Software Engineering (various editions)





Group Discussion

What is Software Engineering?

The Term Software Engineering

What Wikipedia says

D **Software engineering is a systematic engineering approach to software development.**

[Wikipedia]

D **A software engineer is a person who applies the principles of software engineering to design, develop, maintain, test, and evaluate computer software. The term *programmer* is sometimes used as a synonym, but may also lack connotations of engineering education or skills.**

[Wikipedia]

Comments:

- **Do not explain terms with synonyms... software engineering is a kind of software development**
- **Note, software engineering is always systematic**
- **What are the principles of software engineering?**
- **Do not call software engineers programmers! Software engineers might also actually write code but this is not what software engineering is about.**

Some Characteristics

Software engineering copes with constructing large software programs.

Central topic is complexity handling.

Software engineering is a social activity (do not get fooled by the „engineering“). Software engineers talk to people and negotiate the final product in the minefield of conflicting interests.

Software is continuously developed. Even after the original release date.

Productivity and quality are important when developing software.

Software Process Models

“Code and Fix” Model

Approach:

- Write Code
- Detect defects in code
- Fix defects
- Repeat

Problems:

- Unstructured program code
- Not all bugs found

Result:

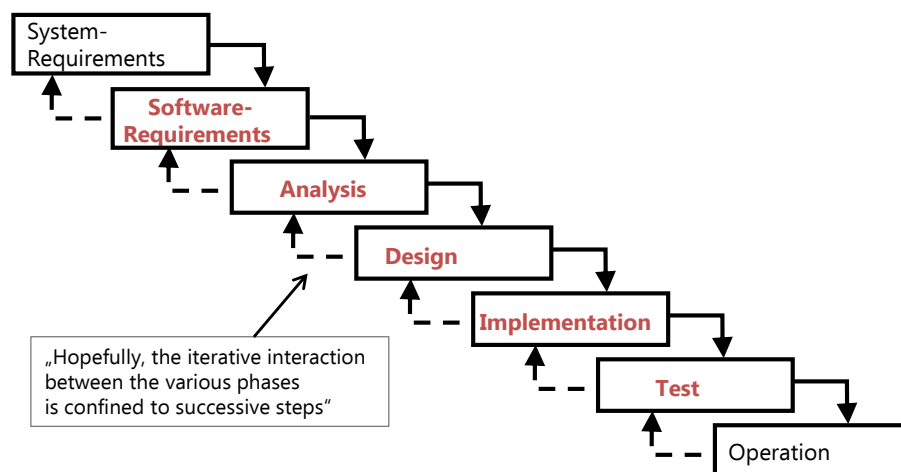
- Customer/user unsatisfied with product



Group Discussion

What can we do better?

Waterfall Model



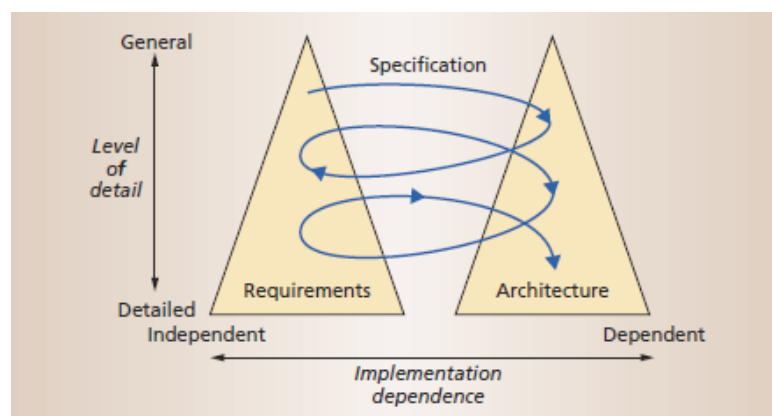
[Royce 1970], later versions by B. Boehm



Group Discussion

What does the Waterfall Model not account for?

Knowledge Gain Explained by Twin Peaks



[Nuseibeh 2001]



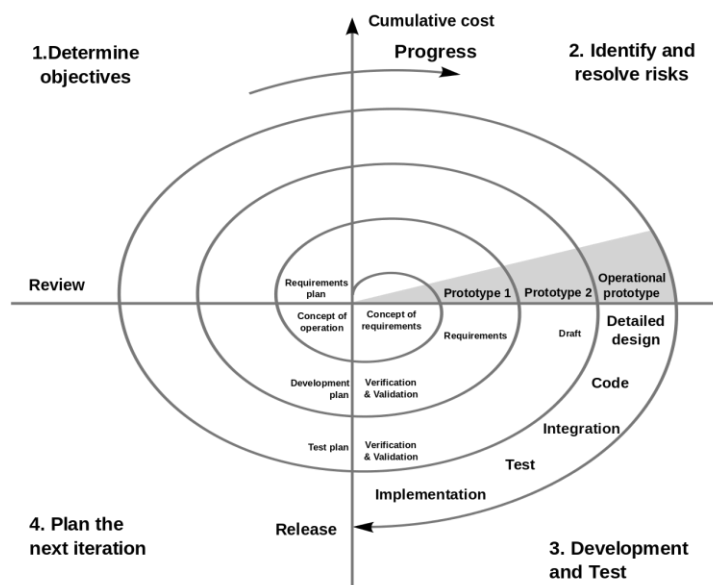
Exercise

Read: Weaving together Requirements and Architectures

(<https://ieeexplore.ieee.org/document/910904>)

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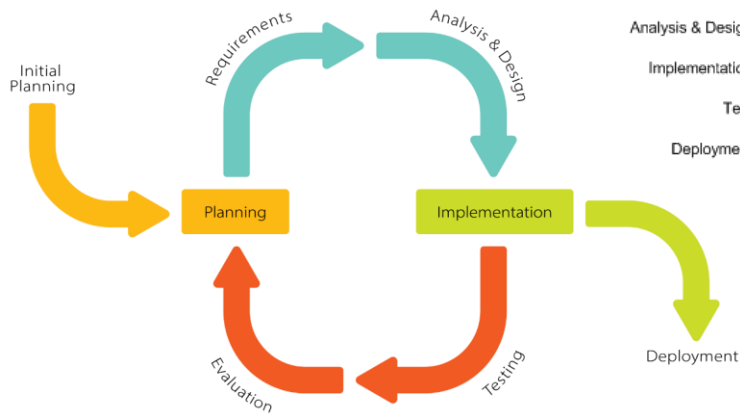
Spiral Model



[Boehm 1988]

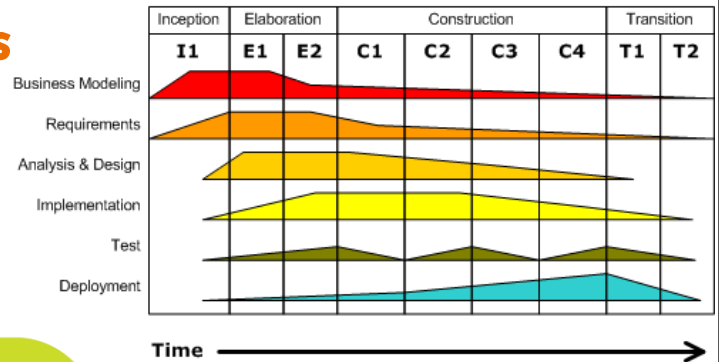
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Iterative Development & Rational Unified Process

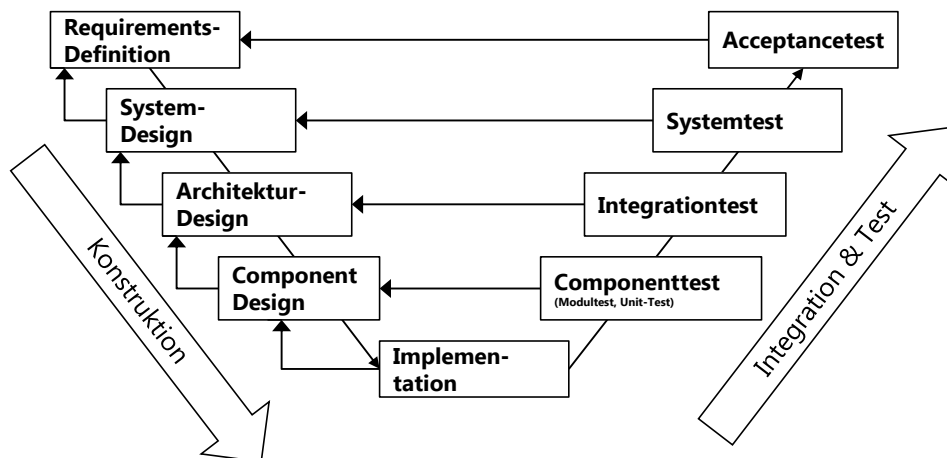


Iterative Development

Business value is delivered incrementally in time-boxed cross-discipline iterations.



V-Model





Group Discussion

There exists a plethora of slightly different V-Models. Why?

The Agile Manifesto

Agile Principles

Individuals and interactions over processes and tools

Working software over comprehensive documentation

Customer collaboration over contract negotiation

Responding to change over following a plan

[Agile Manifesto]

Be Careful

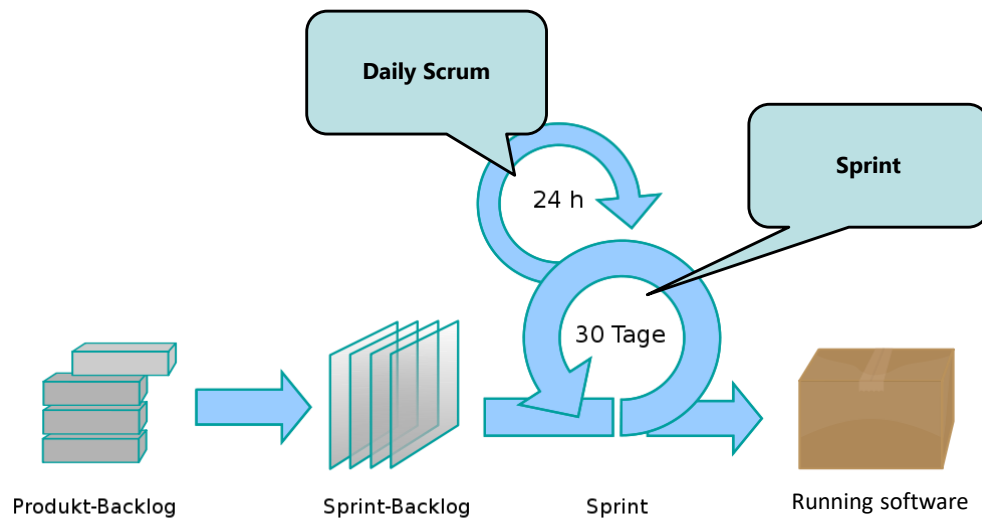
Agile is not unsystematic!

Agile is often used as

“we do not follow a process, we are the cool wicked guys”.

This fails!

Scrum



Three Roles in Scrum

Product Owner

responsible for the product's properties and its economic success; Product Owner structures, details and updates the Product Backlog

Development Team (min 3, max 9)

Scrum Master

moderates meetings, is part of development team but also its leader, is responsible for keeping to the Scrum rules

+ Stakeholders, Clients, Users, Management



Group Discussion

What are the benefits of Agile Methods?

Where is Agile Development applicable?



Exercise

Read the Agile Manifesto

<https://agilemanifesto.org/>



Group Discussion

Does Scrum adhere to the Manifesto?

Major Software Development Phases



Group Discussion

What do all these Software Process Models have in common?

Four Major Phases

Requirements Engineering

Architecture Design

Implementation

Software Quality Assurance

Literature

[Agile Manifesto]	Manifesto for Agile Software Development, 2001.
[Boehm 1988]	B. Boehm: A Spiral Model of software Development and Enhancement. In: IEEE Computer, vol. 21(5), IEEE, 1988.
[Nuseibeh 2001]	B. Nuseibeh: Weaving Together Requirements and Architectures. In: IEEE Computer, vol. 34(3), IEEE, 2001.
[Royce 1970]	W. W. Royce: Managing the Development of Large Software Systems. In: Proc. of IEEE WESCON, IEEE, 1970, pp. 328-388.
[Sommerville]	I. Sommerville: Software Engineering. Pearson Education, various editions, various years.

Questions for Self-Assessment

- **What is software engineering?**
- **What skills does a software engineer need?**
- **Why is “systematic” important?**
- **What is the waterfall model?**
- **What is the v-model?**
- **What are the principles of the agile manifesto?**
- **What is Scrum?**
- **How do traditional and agile development approaches differ? What are advantages? What are disadvantages?**
- **What are the major software development phases?**